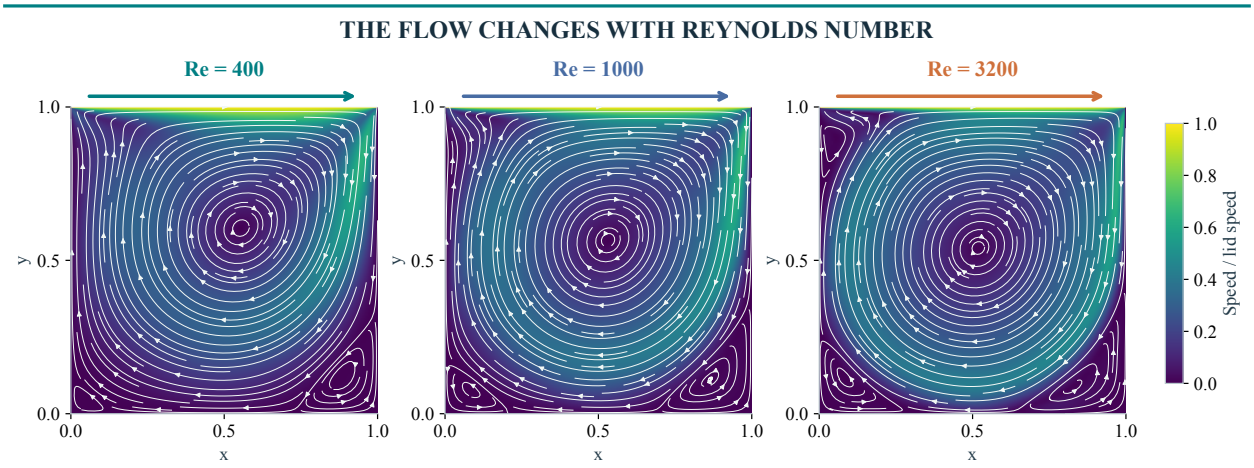
COMPUTATIONAL FLUID DYNAMICS / NUMERICAL STUDY

Navier–Stokes in two dimensions

The lid-driven cavity: approximation, verification, and what the flow reveals



A moving lid generates a primary recirculation, thin wall layers, and secondary corner vortices. This study follows those structures from the incompressible Navier–Stokes equations to a working finite-difference implementation. Each numerical choice is explained together with its cost, its verification, and the limits of the resulting comparison with published data.

Model	Method	Evidence
2-D, viscous, incompressible	Vorticity–streamfunction	Manufactured solutions
Unit square, impulsive lid	Centered space, split time	Two literature references

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Python package ns2d 1.3.0 | Study revision: September 2026

Re = 400, 1000, 3200; three grids per case, fresh integrations, and measured accuracy budgets.

1 Introduction

Predicting the pressure drop in a pipe or the force of air on an aircraft wing requires a mathematical description of how fluids move. The Navier–Stokes equations provide that description for Newtonian fluids: they express conservation of momentum, including the forces due to pressure and viscosity, and are solved together with conservation of mass. In appropriate forms, and coupled to additional physics, they also underpin models of atmospheric and ocean circulation. Their importance comes from this ability to describe many different flows using the same physical balances.

1.1 From viscous forces to a mathematical model

The equations emerged from nineteenth-century attempts to include internal fluid friction in mechanics. In 1822, Claude-Louis Navier introduced viscous forces into the equations of motion; George Gabriel Stokes developed his formulation in 1845. Other researchers contributed to this development, so the name records two major contributions to a longer history [1]. Viscosity accounts for the resistance between neighbouring fluid layers moving at different velocities. It is essential for describing momentum transfer from a solid wall to a fluid, which is precisely the mechanism studied here.

1.2 Why solving the equations is difficult

The main difficulty is the nonlinear transport term $(\mathbf{u} \cdot \nabla)\mathbf{u}$: the velocity being sought also determines how momentum is carried through the domain. A change in the flow therefore changes its subsequent transport. In the incompressible formulation, pressure must adjust throughout the domain to keep $\nabla \cdot \mathbf{u} = 0$, coupling the velocity components through a global constraint.

When inertia is strong relative to viscosity, thin wall layers and small vortical structures can coexist with motion on the scale of the whole domain. A numerical mesh must resolve the relevant spatial scales, while the time step must satisfy both accuracy and stability requirements. Refining the mesh increases the number of unknowns and can require more time steps. These costs, together with complex boundaries, make general flows much harder to compute than the special configurations for which exact solutions are available.

There is also a distinct mathematical question: for the three-dimensional incompressible equations with smooth initial data, do smooth solutions always persist, or can a singularity form? This is the subject of the Clay Millennium problem [2]. Global regularity is known for the corresponding two-dimensional equations on the whole plane or with periodic boundary conditions. The present cavity study concerns numerical approximation in two dimensions; its discontinuous lid velocity introduces boundary singularities of its own.

1.3 How the lid-driven cavity became a benchmark

The lid-driven cavity consists of a square with one moving wall and three fixed walls. In 1966, Burggraf computed this flow up to $\mathrm{Re} = 400$ and examined its primary circulation and secondary corner eddies [3]. Ghia, Ghia and Shin later published multigrid solutions and velocity tables over a wider Reynolds-number range in 1982 [4]; Botella and Peyret supplied a spectral reference at $\mathrm{Re} = 1000$ in 1998 [5]. These studies provide independent numerical comparisons for a geometry that still contains nonlinear transport, wall layers and singular junctions between moving and stationary walls.

1.4 The experiment and its objectives

A square of side L contains a Newtonian fluid of constant kinematic viscosity ν, initially at rest. At time zero the upper wall starts moving to the right at speed U; the other three walls remain stationary. There is no imposed pressure gradient. The moving wall supplies momentum, which spreads through the fluid by advection and viscous diffusion.

This project follows the development of the circulation at $\mathrm{Re} = UL/\nu = 400, 1000, 3200$. It asks how accurately the implementation approximates the equations, how the computed fields change under grid and time refinement, and how they compare with published cavity solutions. Each question requires its own measurements: convergence of an iterative solver alone cannot establish accuracy relative to the continuous equations.

1.5 Scope and evidence

The solver uses a uniform rectangular grid and a scalar streamfunction, so its scope is two-dimensional, simply connected cavities with impermeable walls. The production study uses the unit square and a unit lid. No turbulence model, pressure reconstruction, variable viscosity, or three-dimensional instability analysis is included.

The numerical comparisons use Ghia, Ghia and Shin [4] at three Reynolds numbers, plus Botella and Peyret [5] at $\mathrm{Re} = 1000$. They are *benchmark validation*: comparisons with independently computed solutions of the same mathematical problem, not experimental validation of a real apparatus.

Question	Evidence supplied	Limit of the evidence
Implementation accuracy	Smooth manufactured solutions; space and time refinement	Does not establish the same global order at singular lid corners
Algebraic correctness	Poisson residual, divergence and circulation checks	Identities can hold on an inaccurate flow field
Benchmark agreement	Centerlines and primary vortex versus two papers	References have their own discretisation errors
Steady convergence	Update and steady-PDE residuals; ten-time-unit continuation	Measured drift is not a rigorous remaining-error bound
Discretisation sensitivity	Three grids per Re; transient time refinement	Richardson estimates assume a leading power-law error

1.6 A guide to the report

The report first derives the model, spatial stencils, wall treatment, and time step. It then tests accuracy with a smooth manufactured solution, interprets the cavity fields, and compares numerical references. The final section maps the mathematics to the software and gives reproduction commands.

All nine production states were integrated afresh from rest with the current numerical kernel. Each record includes the case parameters, start and finish timestamps, dependency versions, numerical-source fingerprint, runner fingerprint, and hashes of the saved fields and cache. The results builder rejects stale, altered, or unconverged production states. The manufactured verification and cavity sensitivity experiments are separately reproducible.

2 From velocity and pressure to two scalar fields

2.1 Nondimensional equations

Scale lengths by L, velocities by U, time by L/U, pressure by ρU^2, vorticity by U/L, and streamfunction by UL. Dropping the dimensionless asterisks gives

$$\partial_t \mathbf{u} + (\mathbf{u} \cdot \nabla)\mathbf{u} = -\nabla p + \frac{1}{\mathrm{Re}}\Delta \mathbf{u}, \tag{1}$$

$$\nabla \cdot \mathbf{u} = 0, \qquad \mathrm{Re} = \frac{UL}{\nu}, \qquad \mathbf{u} = (u, v). \tag{2}$$

Thus Re measures the ratio of inertial to viscous effects. Time $t = 1$ is one lid travel time across the cavity.

Set

$$u = \psi_y, \qquad v = -\psi_x, \qquad \omega = v_x - u_y = -\Delta\psi. \tag{3}$$

The divergence vanishes because mixed derivatives commute. Taking the curl of (1) eliminates pressure. In two dimensions the vortex-stretching term vanishes, leaving

$$\boxed{\omega_t + \psi_y \omega_x - \psi_x \omega_y = \mathrm{Re}^{-1}\Delta\omega, \qquad -\Delta\psi = \omega.} \tag{4}$$

The convection sign follows directly from $u\omega_x + v\omega_y$. A positive lid speed drives a clockwise primary circulation: the primary streamfunction is negative in this convention.

2.2 Initial and boundary data

On $\Omega = (0, 1)^2$, the velocity is

$$(u, v) = \begin{cases} (1, 0), & y = 1, \quad 0 < x < 1, \\ (0, 0), & \text{on the remaining open wall segments.} \end{cases} \tag{5}$$

Impermeability makes ψ constant along the connected boundary; choose that constant as zero. Tangential no-slip supplies the normal derivative of ψ, which will determine wall vorticity. The interior initial condition is rest: $\psi = \omega = 0$ at $t = 0$.

The top corners are incompatible boundary junctions: the lid prescribes $u = 1$ while each side prescribes $u = 0$. The impulsive start is also nonsmooth in time. The solver does not smooth the lid, because that would define a different benchmark. Stored corner velocities follow the side-wall convention; wall vorticity at these singular points has no grid-independent point value.

Interpretation. The scalar formulation makes incompressibility straightforward and removes pressure from the solve. The difficulty moves to the wall vorticity. A velocity–pressure projection method would be more natural for extensions to three dimensions; it is not required for this two-dimensional study.

3 Approximating space

3.1 Grid and finite differences

Use $N_x \times N_y$ points, including the boundaries, with $x_i = ih_x$, $y_j = jh_y$, $h_x = 1/(N_x - 1)$ and $h_y = 1/(N_y - 1)$. Arrays use field[i,j]: x is the first axis. A 129^2 mesh has $h = 1/128$, not $1/129$.

For any smooth scalar field q, define

$$D_x q_{ij} = \frac{q_{i+1,j} - q_{i-1,j}}{2h_x}, \qquad D_y q_{ij} = \frac{q_{i,j+1} - q_{i,j-1}}{2h_y}, \tag{6}$$

$$\Delta_h q_{ij} = \frac{q_{i+1,j} - 2q_{ij} + q_{i-1,j}}{h_x^2} + \frac{q_{i,j+1} - 2q_{ij} + q_{i,j-1}}{h_y^2}. \tag{7}$$

Taylor expansion yields, for example, $D_x q = q_x + (h_x^2/6) q_{xxx} + O(h_x^4)$. The interior spatial truncation error is second order when the required derivatives remain bounded.

The transport term is evaluated in advective form:

$$C_h(\psi, \omega)_{ij} = (D_y \psi)_{ij} (D_x \omega)_{ij} - (D_x \psi)_{ij} (D_y \omega)_{ij}. \tag{8}$$

It uses the current streamfunction and vorticity, including the wall values needed by the first interior row of centered differences.

3.2 Why use these approximations?

Choice	Benefit	Cost or alternative
Uniform grid	Transparent stencils; separable Poisson operator	Many nodes are spent in the smooth core rather than thin wall layers
Centered convection	Second-order consistency without a first-order upwind diffusion term	No general monotonicity guarantee; under-resolved oscillations are possible
Five-point Laplacian	Compact SPD Dirichlet system	Second-order spatial accuracy; smoothness assumptions fail at lid corners
Advective Jacobian	Direct implementation of $u\omega_x + v\omega_y$	Does not inherit the energy/enstrophy properties of an Arakawa Jacobian

Arakawa's construction [7] is an important alternative when preserving the quadratic invariants of inviscid two-dimensional advection is a priority. This implementation uses the simpler centered product and makes no claim of exact energy or enstrophy conservation. In the viscous driven cavity, the moving boundary supplies energy and diffusion dissipates it, so the energy itself should not be constant during the transient.

Interpretation. Formal order describes the local approximation of smooth fields. Resolution and boundary regularity determine whether that order is visible in a particular cavity observable.

4 How the moving wall creates vorticity

The Poisson equation needs one boundary value for ψ. With $\psi = 0$ already imposed, the no-slip information is transferred to ω. At the top wall, away from the corners, Taylor expansion gives

$$\psi(x, 1 - h_y) = -h_y u_T + \frac{h_y^2}{2}\psi_{yy}(x, 1) + O(h_y^3). \tag{9}$$

Because ψ is constant along that wall, $\psi_{xx} = 0$ there. Hence $\psi_{yy} = -\omega$ and

$$\boxed{\omega_{i,N_y-1} = -\frac{2\psi_{i,N_y-2}}{h_y^2} - \frac{2u_T(x_i)}{h_y}.} \tag{10}$$

This is the Thom wall formula. A lid moving to the right produces negative vorticity next to an initially motionless interior, as expected for clockwise rotation.

For general tangential wall velocities, the other formulas are

$$\omega_{i,0} = -2\psi_{i,1}/h_y^2 + 2u_B(x_i)/h_y, \tag{11}$$

$$\omega_{0,j} = -2\psi_{1,j}/h_x^2 - 2v_L(y_j)/h_x, \tag{12}$$

$$\omega_{N_x-1,j} = -2\psi_{N_x-2,j}/h_x^2 + 2v_R(y_j)/h_x. \tag{13}$$

In the standard cavity, $u_T = 1$ and the other velocities vanish.

4.1 Local order is not global order

Dividing the Taylor remainder by h^2 leaves an $O(h)$ wall truncation error. That does not by itself establish a first-order global streamfunction error: the wall error enters a coupled elliptic and transport problem. The observed global order must be measured. Section 7 finds approximately second-order streamfunction convergence on the selected smooth manufactured field. This is consistent with the role of local vorticity formulas discussed by E and Liu [6]; it is not a universal guarantee for a singular cavity.

4.2 What the corner convention means

No interior five-point Laplacian or centered convection stencil reaches a corner. Changing the four stored corner vorticities therefore leaves the computed interior unchanged. This implementation stores the sum of the two adjacent wall formulas at each corner. That convention makes a particular trapezoidal circulation identity exact; it is bookkeeping for integration, not a physical evaluation of the singularity.

The displayed wall vorticity is lagged by one step: it was constructed from ψ^n, whereas the stored interior is ω^{n+1}. This distinction matters when checking circulation or interpreting transient wall values.

Interpretation. The wall formula is the mechanism that drives this solver. A verification field with zero tangential wall velocities cannot test its velocity-dependent terms.

5 Advancing time and solving the systems

5.1 One semi-implicit step

Let $L = -\Delta_h$ on the interior with homogeneous Dirichlet conditions. Let b^n contain the Thom wall ring from ψ^n, with zero interior. The algorithm performs

$$(I + \Delta t \mathrm{Re}^{-1} L)\omega_I^{n+1} = \omega_I^n - \Delta t C_h(\psi^n, \omega^n) + \Delta t \mathrm{Re}^{-1}(\Delta_h b^n)_I + \Delta t f_I^{n+1}, \tag{14}$$

$$L\psi_I^{n+1} = \omega_I^{n+1}, \qquad \psi^{n+1}|_{\partial\Omega} = 0. \tag{15}$$

The source f vanishes for the cavity and is used only for manufactured solutions. In exact-wall verification, the diffusion solve instead uses the known wall vorticity at t^{n+1}. Convection always uses the current-time wall values. Explicit convection, backward Euler diffusion, and the lagged wall treatment make this a *first-order* time scheme.

The wall-ring term in (14) is simply Dirichlet-column elimination. For example, the node below the lid receives $+(\Delta t/\mathrm{Re}h_y^2)\omega^n_{i,N_y-1}$ on its right-hand side. No nonlinear iteration is performed.

5.2 Why the linear solves are well behaved

The Dirichlet negative Laplacian is symmetric positive definite. Therefore both L and $I + \Delta t \mathrm{Re}^{-1} L$ are suitable for conjugate gradient. On this uniform rectangle, a faster default is available: the type-I discrete sine transform diagonalises L. Its eigenvalues are

$$\lambda_{pq} = \frac{4}{h_x^2}\sin^2\frac{p\pi}{2(N_x-1)} + \frac{4}{h_y^2}\sin^2\frac{q\pi}{2(N_y-1)}. \tag{16}$$

For $(\alpha I + \beta L)z = r$, transform r, divide each coefficient by $\alpha + \beta\lambda_{pq}$, and inverse transform. The solve is exact up to floating-point error for the *discrete* system, at $O(M \log M)$ cost for $M = (N_x - 2)(N_y - 2)$ unknowns.

Two alternative backends serve as checks: sparse LU, factored once and reused, and conjugate gradient with an incomplete-Cholesky IC(0) preconditioner. The latter is SPD; a generic incomplete-LU preconditioner would not provide that required property. Iterative solutions agree to the requested tolerance, not necessarily bit for bit.

5.3 When is a run steady?

The driver monitors the interior update rate

$$R_n = \frac{\|\omega_I^{n+1} - \omega_I^n\|_\infty}{\Delta t}. \tag{17}$$

The production stopping threshold is 10^{-6}, tested every 50 steps. It is an absolute update-rate criterion, not an estimate of spatial error. A fixed-step run stops at the first time at or beyond its requested T; the final step is not shortened and can overshoot by less than Δt. The production time caps are safeguards, not declarations that a flow has settled. Every reported production state must meet the criterion. The accuracy study then checks a further ten time units with both Δt and $\Delta t/2$.

6 Stability: the interior and the walls

Implicit diffusion removes the usual explicit interior diffusion restriction, but it does not make the full algorithm unconditionally stable. For constant advection velocity (a, b) on an infinite uniform grid, a Fourier mode has amplification factor

$$G = \frac{1 - i\Delta t A}{1 + \nu \Delta t B}, \quad A = \frac{a \sin\theta_x}{h_x} + \frac{b \sin\theta_y}{h_y}, \quad B = \frac{4\sin^2(\theta_x/2)}{h_x^2} + \frac{4\sin^2(\theta_y/2)}{h_y^2}, \tag{18}$$

where here $\nu = 1/\mathrm{Re}$. Since $A^2 \le (a^2 + b^2)B$, a sufficient bound for this *linear interior* analysis is

$$\Delta t \le \frac{2\nu}{a^2 + b^2}. \tag{19}$$

This argument assumes constant coefficients and does not include the cavity walls or nonlinear interactions.

6.1 A boundary experiment on the implemented step

Turn convection off, set all wall velocities to zero, and reconstruct ψ from each interior basis vector of ω. One step gives a linear map $\omega_I^{n+1} = M\omega_I^n$. The module ns2d.stability builds M using the production step and computes its spectral radius. With $d = \nu\Delta t/h^2$, the measured crossings of $\rho(M) = 1$ are

Grid	17^2	25^2	33^2
Critical d	1.58085	1.54647	1.53199

The finite-grid trend motivates using $d_{\max} = 1.5$ with a safety factor. Prescribing a fixed zero wall-vorticity ring removes this feedback mechanism. These tests diagnose a boundary restriction; they do not prove a sharp nonlinear threshold or a theorem for stretched rectangular grids.

The default helper combines the three safeguards

$$\Delta t = s \min\left(\frac{h_{\min}}{U_*}, \frac{2}{\mathrm{Re}U_*^2}, 1.5\mathrm{Re}h_{\min}^2\right), \qquad s = 0.5, \tag{20}$$

where U_* is an estimated maximum speed. The cavity constructor uses the configured lid-speed magnitude. A new flow configuration should be checked with a smaller step; the helper is not an adaptive stability controller.

6.2 A mesh diagnostic is not a stability theorem

The cell Reynolds number is $\mathrm{Re}_h = \mathrm{Re}Uh$ (unit lid: $\mathrm{Re}_h = \mathrm{Re}h$). The familiar $\mathrm{Re}_h \le 2$ condition concerns monotonicity of a one-dimensional steady centered advection–diffusion stencil. It is a useful warning about resolution, not a universal cavity accuracy or temporal-stability criterion. On the 257^2 grids here it is 1.56, 3.91, and 12.5. Reducing Δt cannot substitute for resolving a thin wall layer.

7 Verification with a known smooth solution

Choose the manufactured streamfunction

$$\psi_e = \sin(\pi x)\sin(\pi y)e^{-t} + \tfrac{1}{2}\sin(2\pi x)\sin(2\pi y)e^{-2t}, \qquad \omega_e = -\Delta\psi_e. \tag{21}$$

It vanishes on the boundary but has nonzero normal derivatives there, so the wall velocities exercise the moving-wall terms. Its two Laplacian eigenvalues differ; consequently ω_e is not proportional to ψ_e and convection does not vanish identically.

Symbolic differentiation defines the continuous forcing

$$f = \partial_t\omega_e + (\partial_y\psi_e)(\partial_x\omega_e) - (\partial_x\psi_e)(\partial_y\omega_e) - \mathrm{Re}^{-1}\Delta\omega_e. \tag{22}$$

This is a continuous manufactured source, not a discrete source chosen to make the numerical answer exact. All tests use $\mathrm{Re} = 100$ and start from the exact continuous fields; the initial pair need not satisfy the discrete Poisson relation exactly.

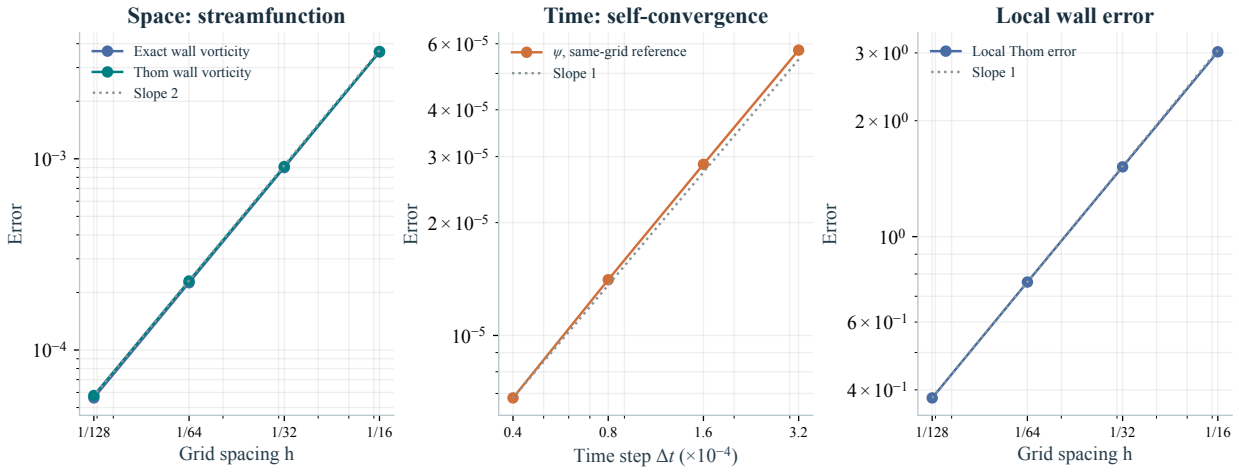
7.1 Errors and observed order

Use the composite trapezoidal norm

$$\|e\|_{2,h} = \left(h_x h_y \sum_{ij} \sigma_{ij} e_{ij}^2\right)^{1/2}, \qquad p = \frac{\log(E_h/E_{h/2})}{\log 2}, \tag{23}$$

where σ is 1 inside, $1/2$ on an edge, and $1/4$ at a corner. The four inert vorticity corners are excluded from both vorticity norms without renormalising other weights; the streamfunction norms use the full grid.

- **Space:** $N = 17, 33, 65, 129$, fixed $\Delta t = 10^{-5}$, $T = 0.005$. Run both exact wall vorticity and the production Thom wall treatment.
- **Time:** fixed 65^2 grid, $T = 0.032$, with $\Delta t = 3.2, 1.6, 0.8, 0.4 \times 10^{-4}$. Compare to a same-grid solution with $\Delta t_{\mathrm{ref}} = 2.5 \times 10^{-6}$ to reduce the spatial-error floor.
- **Joint refinement:** $N = 17, 33, 65$, $T = 0.04$ and $\Delta t = 4 \times 10^{-4}(h/h_{17})^2$.



Spatial streamfunction error, temporal self-convergence, and local wall truncation error. Dashed guides show the expected slopes.

7.2 What the convergence measurements say

Measurement	Successive observed orders
Spatial ψ, exact wall ω, L^2	2.007, 2.002, 2.001
Spatial ψ, Thom wall ω, L^2	1.999, 1.989, 1.989
Spatial ω, Thom wall, L^2	1.598, 1.819, 2.071
Spatial ω, Thom wall, maximum norm	1.080, 1.349, 1.931
Temporal ψ, same-grid reference, L^2	1.011, 1.023, 1.047
Joint refinement ψ, L^2	2.002, 1.997

The streamfunction measurements support second-order spatial consistency for this smooth test and first-order time accuracy. Vorticity converges less uniformly, particularly in the wall-sensitive maximum norm. Reporting its full sequence is more informative than selecting only the finest-grid order. The finite temporal reference introduces a small bias: for an ideal first-order error, differences scale like $\Delta t - \Delta t_{\mathrm{ref}}$, so ratios can give orders slightly above one.

Two additional checks challenge the verification setup. An asymmetric field, $\psi_e = \sin(\pi x)\sin(2\pi y)e^{-t} + 0.3\sin(3\pi x)\sin(\pi y)e^{-2t}$, is integrated to $T = 0.05$ on $17^2, 33^2, 65^2$ grids: its streamfunction orders are approximately 1.97 and 2.00. Halving Δt in the original finest-grid spatial test changes its streamfunction L^2 error by less than 0.003%. These checks reduce concern that symmetry, the short observation time, or temporal contamination produced the reported spatial slope. All errors remain available in the JSON evidence.

7.3 Algebraic checks and their interpretation

For reconstructed velocity, the centered mixed differences cancel:

$$D_x u + D_y v = D_x D_y \psi - D_y D_x \psi = 0. \tag{24}$$

The cancellation also holds next to impermeable straight walls with the implemented boundary velocities. The Poisson residual measures a different identity:

$$r_P = \|\Delta_h \psi + \omega_I\|_\infty. \tag{25}$$

Both quantities are recomputed for every fresh cavity state and exported in data/validation.json; the Poisson audit requires $r_P < 10^{-8}$.

For a vorticity field constructed consistently from one ψ, the wall formulas, interior Laplacian and sum corner convention telescope under the trapezoidal rule:

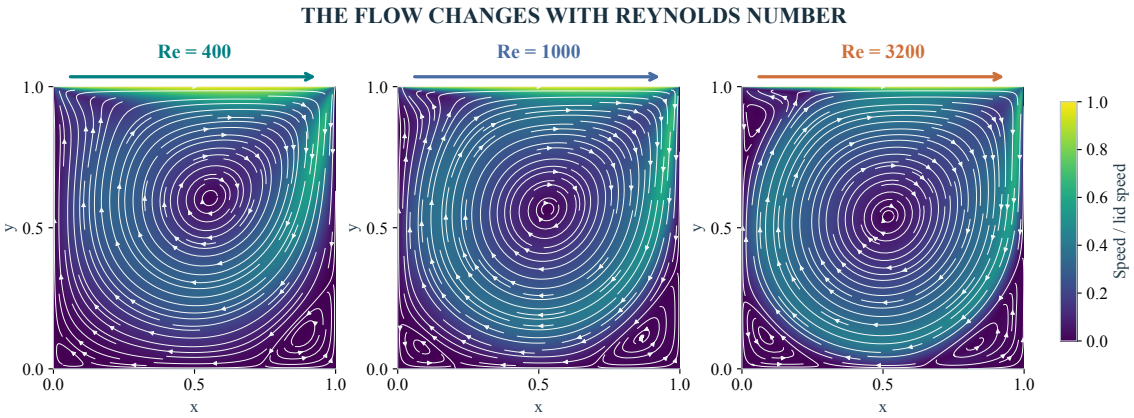
$$h_x h_y \sum_{ij} \sigma_{ij}\omega_{ij} = \oint_{\partial\Omega} \mathbf{u}\cdot d\boldsymbol{\ell} = -1. \tag{26}$$

Here the contour orientation is counterclockwise. The implementation of circulation_identity constructs its own interior $-\Delta_h\psi$; it checks the discrete formulas and quadrature, not the independently stored Poisson solution. The latter is checked by r_P. Direct integration of stored ω also includes its one-step wall lag.

Tests additionally compare DST, LU and CG, check invariance to inert corner values, and compare a checkpoint restart with uninterrupted stepping. These checks constrain different failure modes. None replaces the refinement study.

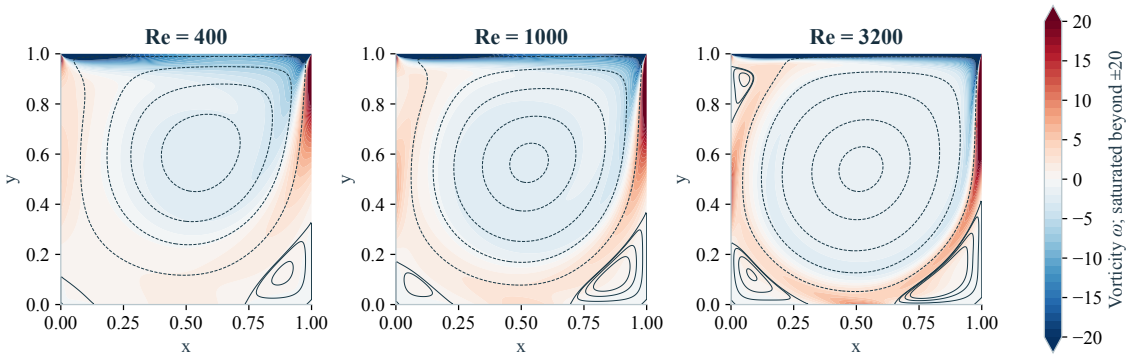
Interpretation. The verification evidence is strong for the selected smooth problem. It should not be reinterpreted as proof of second-order accuracy for every quantity in the singular lid-driven cavity.

8 What the cavity flow reveals



Velocity streamlines over speed on 257^2 grids. The common colour scale is 0–1 in lid-speed units. These are instantaneous streamlines, not particle trajectories through the transient.

The lid first creates a shear layer. Fluid moves right beneath it, turns down the right wall, returns along the bottom, and rises along the left wall. The primary circulation is clockwise. As Re increases, viscous transport becomes weaker relative to advection, wall gradients sharpen, and the primary vortex moves toward the cavity centre. Counter-rotating corner cells become more prominent; at Re = 3200 a top-left cell is visible.

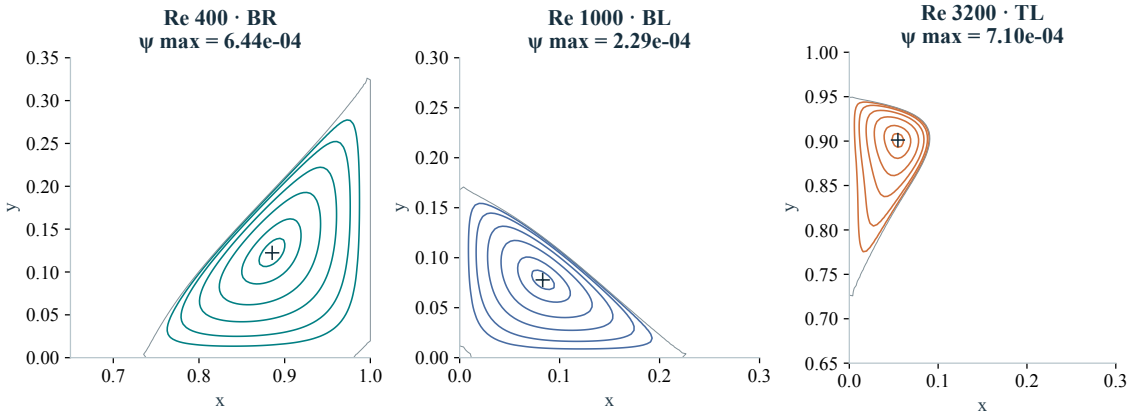


Vorticity with streamfunction contours. The same range $[-20, 20]$ is used in all panels. Saturated colours hide larger magnitudes; the full fields, including the lid-corner values, are not clipped in the data.

The vorticity field exposes the wall layers more clearly than a streamline plot. Fine structure near walls is also where centered advection on a coarse mesh is most vulnerable. A smooth-looking streamline plot alone is therefore an inadequate accuracy check.

8.1 Secondary circulation at the corners

The primary vortex carries most of the visible motion, but it is not the only recirculation in the cavity. Near a corner the main flow can turn away from both walls and leave a small region rotating in the opposite direction. In the present sign convention those counter-rotating cells have positive streamfunction. They occupy a much smaller range of ψ than the primary vortex, so a uniform set of contour levels over the whole cavity can miss them.



Selected positive-streamfunction corner cells. Contour levels are fractions of each local maximum, so colours do not compare vortex strength across panels. Crosses mark local quadratic-fit extrema.

Case	Selected cell	$\psi_{\max}$	x	y
400	Bottom right	6.4449e-04	0.88541	0.12238
1000	Bottom left	2.2885e-04	0.08308	0.07779
3200	Top left	7.1028e-04	0.05472	0.90119

Selected extrema of the present 257^2 fields, found by local quadratic fits. They are computed quantities, not reference values. Scientific notation "e" denotes powers of ten. Grid sensitivity of the bottom-right cell is included in the accuracy study.

Each zoom uses levels at 5, 15, 35, 60, 85 and 97 percent of that cell's maximum. This reveals its shape while making the normalisation explicit. A cross marks the fitted extremum; the zero contour separates positive streamfunction from the clockwise primary circulation. Contour appearance must therefore not be used to compare strengths without the numbers above.

These eddies are a useful resolution test. Their amplitudes are orders of magnitude smaller than the primary streamfunction, and their wall gradients are sampled by relatively few nodes on a uniform mesh. Good agreement in primary-vortex strength does not imply equal accuracy for every small cell. The current search finds the strongest positive cell in each corner quadrant; it is not an exhaustive detector of nested higher-order vortices.

Interpretation. A figure can reveal flow structure only at the scales it displays. Global plots explain the circulation; local plots and explicit contour levels make the smaller features interpretable.

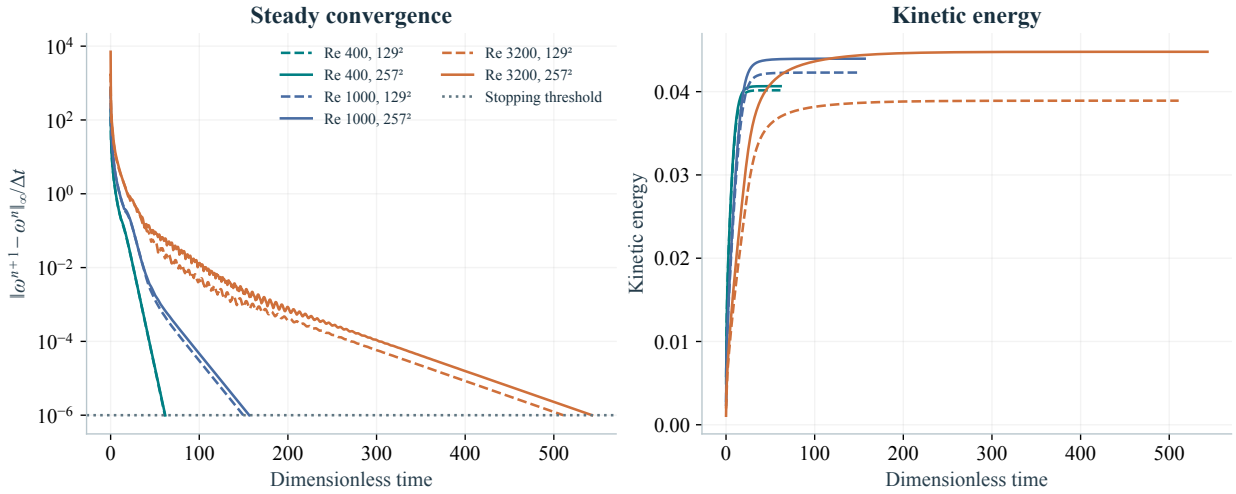
9 Convergence and comparison with Ghia

The reported primary vortex is a local quadratic fit around the grid minimum of ψ. It reduces node-location quantisation but is not a new high-order flow solution. The relative discrepancy in its strength is $100|\psi_{\min} - \psi_{\mathrm{ref}}|/|\psi_{\mathrm{ref}}|$.

Re	Grid	Final t	R	Steady?	$\psi_{\min}$	Difference (%)
400	65^2	61.12	9.80e-07	yes	-0.109052	4.26
400	129^2	61.37	9.68e-07	yes	-0.112776	0.99
400	257^2	61.62	9.88e-07	yes	-0.113689	0.19
1000	65^2	140.00	9.97e-07	yes	-0.105252	10.75
1000	129^2	149.70	9.98e-07	yes	-0.115473	2.08
1000	257^2	156.40	9.97e-07	yes	-0.118074	0.12
3200	129^2	510.66	1.00e-06	yes	-0.109979	8.64
3200	193^2	534.75	1.00e-06	yes	-0.116413	3.29
3200	257^2	543.11	1.00e-06	yes	-0.118734	1.37

Reference strengths: -0.113909, -0.117929, and -0.120377 for Re = 400, 1000, 3200 respectively (Ghia, Table V). Scientific notation "e" denotes a power of ten.

All nine integrations meet the same 10^{-6} stopping threshold. The final times differ because settling depends on both Reynolds number and grid. The following accuracy study reports the change during a further ten time units and explicitly compares earlier $t = 150$ or $t = 300$ snapshots with the converged fine-grid states. A residual threshold alone is not an error bound.

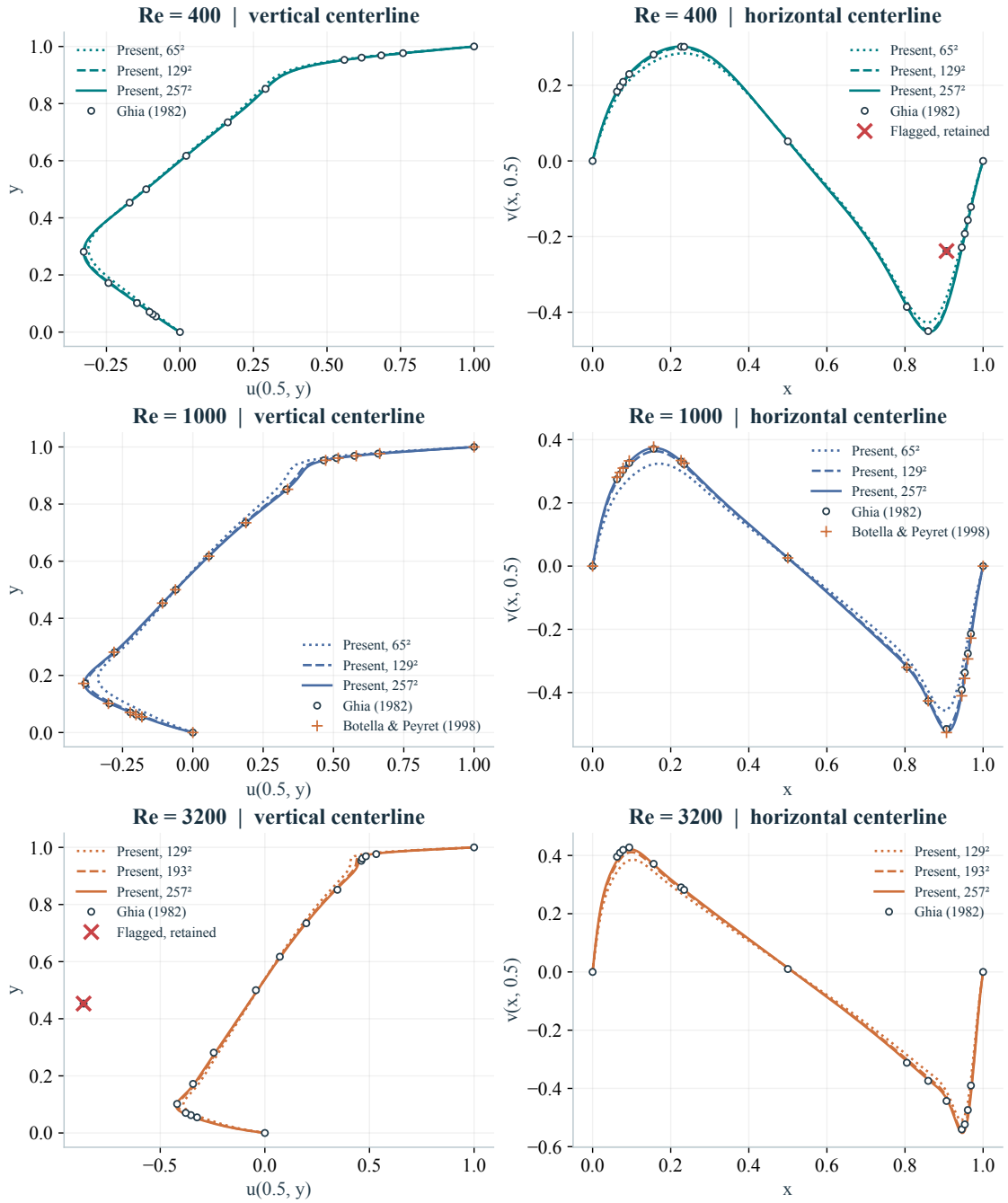


Recorded interior update residuals and kinetic energy on the 129^2 and 257^2 grids. The threshold is 10^{-6}. Energy is a global diagnostic and can appear almost constant while the maximum update residual remains above the threshold.

9.1 What grid refinement supports

The primary-vortex comparison improves with refinement. Other observables need individual checks: at Re = 400, for example, the unflagged vertical profile discrepancy does not decrease between 129^2 and 257^2. The next section measures differences between our own grids, independently of Ghia's values, and reports conditional Richardson estimates rather than equating benchmark agreement with continuum accuracy.

9.2 Centerline profiles, with every reference point visible



Computed profiles are drawn from the full grid; markers are published stations. Red crosses flag suspect published values but retain them in the headline RMS. The orange crosses at Re = 1000 show the reflected Botella–Peyret reference. The Re = 1000 and 3200 curves are states that meet the common steady criterion. Their remaining drift is measured in the accuracy study.

9.3 Reference anomalies and reproducible metrics

For $m = 17$ reference stations, the profile error is

$$E_{u,\mathrm{RMS}} = \sqrt{\frac{1}{m}\sum_{k=1}^{m}\left[u_h(0.5, y_k) - u_{\mathrm{ref},k}\right]^2}, \tag{27}$$

and likewise for v. Linear interpolation evaluates the numerical fields at the printed coordinates. The zero-velocity endpoints are included; this is an absolute dimensionless velocity error, not a percentage.

The original paper was inspected at Tables I and II (pp. 398–399) and Table V (pp. 408–409). Two anomalous stations affect this study:

- At Re = 3200, Table I prints $u = -0.86636$ at $y = 0.4531$. It is retained. Replacing it by -0.086636 based on the current computation would use the answer under test to alter the reference.
- At Re = 400, Table II prints $v = -0.23827$ at $x = 0.9063$. It is retained; a second RMS shows the effect of excluding that one point.

These are flags for interpretation, not claims of an official erratum. One genuine local transcription error was corrected: the Re = 1000 vertical velocity at $x = 0.9063$ is printed as -0.51550, not -0.51500.

Re	Grid	u all	u unflagged	v all	v unflagged
400	65^2	1.252e-02	1.252e-02	3.254e-02	1.399e-02
400	129^2	2.573e-03	2.573e-03	3.502e-02	2.143e-03
400	257^2	8.930e-04	8.930e-04	3.635e-02	2.254e-03
1000	65^2	3.795e-02	3.795e-02	4.062e-02	4.062e-02
1000	129^2	8.130e-03	8.130e-03	5.309e-03	5.309e-03
1000	257^2	1.806e-03	1.806e-03	6.614e-03	6.614e-03
3200	129^2	1.946e-01	3.301e-02	3.124e-02	3.124e-02
3200	193^2	1.916e-01	1.421e-02	1.309e-02	1.309e-02
3200	257^2	1.911e-01	1.130e-02	1.306e-02	1.306e-02

Full and secondary RMS errors. “Unflagged” uses 16 stations in the affected component and 17 otherwise. All station values, interpolated results and flags are exported in data/centerline_comparisons.csv.

The primary-vortex reference is not one uniform 129^2 data set. Table V explicitly lists 257^2 at Re = 400 and 129^2 at Re = 1000, 3200. The printed Re = 3200 centre $x = 0.5165$ is retained even though it is off the listed mesh. We do not infer or repair a reference by rounding it to a preferred grid.

Interpretation. Raw reference data and a transparent sensitivity calculation make an anomalous comparison inspectable. A smaller score obtained after changing the source values would be much harder to interpret.

10 An independent spectral reference at Re = 1000

Botella and Peyret [5] use Chebyshev collocation and subtract leading corner singularities. This provides a useful second numerical method against which to compare the finite-difference result. Their Tables 6, 9 and 10 supply the primary vortex and both centerlines.

10.1 Reconciling the lid direction

Their lid moves left. Reflect the field about $x = 1/2$ to compare with the right-moving lid used here:

$$x_R = 1 - x_L, \qquad y_R = y_L, \tag{28}$$

$$u_R(x_R, y) = -u_L(1 - x_R, y), \qquad v_R(x_R, y) = v_L(1 - x_R, y), \tag{29}$$

$$\psi_R(x_R, y) = -\psi_L(1 - x_R, y). \tag{30}$$

In particular, v does *not* change sign. After reflection, the reference primary vortex is $\psi = -0.1189366$ at $(0.5308, 0.5652)$. The code preserves the left-lid tabulation and applies the transformation explicitly when constructing the comparison.

Quantity	Present, 257^2	Ghia	Botella–Peyret
$\psi_{\min}$	-0.1180744	-0.1179290	-0.1189366
Vortex x	0.53107	0.5313	0.5308
Vortex y	0.56560	0.5625	0.5652
Strength discrepancy	N/A	0.123%	0.725%
u RMS discrepancy	N/A	1.806e-03	2.724e-03
v RMS discrepancy	N/A	6.614e-03	3.113e-03

10.2 What this changes in the interpretation

Agreement with Ghia is close, but it is not a continuum error bar. The streamfunction is closer to Ghia, whereas the vertical profile and vortex ordinate are closer to the spectral reference. The comparison reveals the size and direction of differences between reference methods. It also shows why one small percentage cannot summarise every feature of the flow.

The present field now meets the steady threshold. A separate continuation study measures its small remaining drift. The spectral reference reports higher accuracy in its own convergence study, but benchmark discrepancies still combine errors of both methods. Our three-grid study supplies a distinct, explicitly conditional estimate of our discretisation uncertainty.

Interpretation. A useful numerical portfolio explains why an answer should be trusted, how it compares with more than one method, and where the evidence stops. These discrepancies are part of the result.

11 Separating grid, time and settling effects

All measurements in this section are generated by scripts/run_accuracy.py and retained in data/accuracy.json. They use the same cavity equations, wall treatment and interior stencils as the production study.

11.1 A three-grid estimate independent of the benchmarks

For Re = 400, 1000 use $N = 65, 129, 257$. For Re = 3200 use $N = 129, 193, 257$, avoiding the very coarse 65^2 grid in the order estimate. Each state starts from rest and reaches the same steady threshold. Write $h_c > h_m > h_f$ for the coarse, medium and fine spacings. For a scalar observable q, assume a leading error model

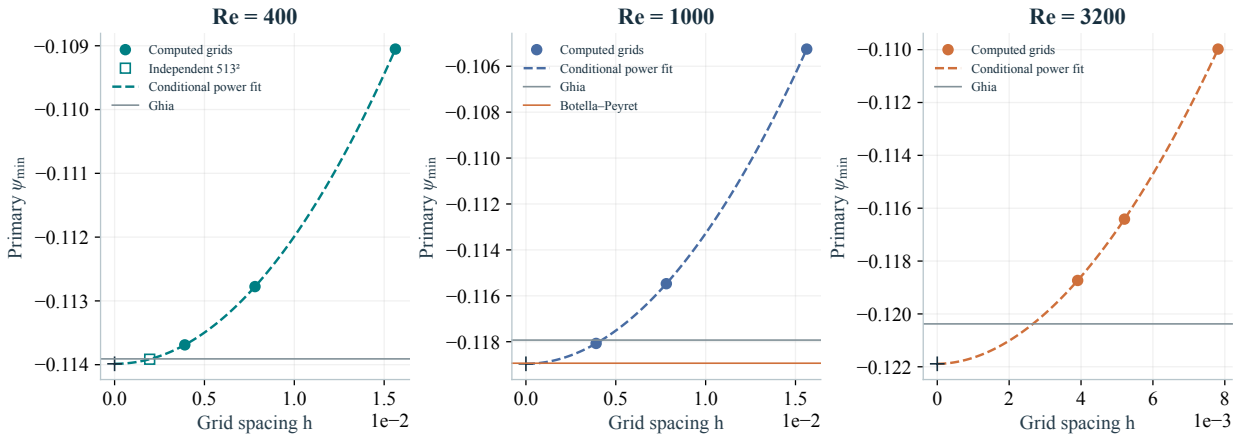
$$q(h) = q_0 + Ch^p, \qquad \frac{q_c - q_m}{q_m - q_f} = \frac{h_c^p - h_m^p}{h_m^p - h_f^p}. \tag{31}$$

The latter equation is solved for p using the actual spacings; no equal-ratio formula is imposed on the Re = 3200 sequence. It gives

$$q_0^{\text{est}} = q_f + \frac{q_f - q_m}{(h_m/h_f)^p - 1}, \qquad U_h = 1.25\frac{|q_f - q_m|}{(h_m/h_f)^p - 1}. \tag{32}$$

The corresponding relative grid-convergence index is $100U_h/|q_f|$, following the three-grid safety factor of Celik et al. [8]. Non-monotone sequences, roundoff-level differences, and sequences with no positive fitted order in $(0.01, 10)$ receive no extrapolation.

Re	N values	Fitted p	$\psi_{\min,0}^{\text{est}}$	GCI (%)
400	65, 129, 257	2.027	-0.1139863	0.327
1000	65, 129, 257	1.974	-0.1189628	0.941
3200	129, 193, 257	1.917	-0.1218878	3.321



Computed primary streamfunctions and conditional fits. The point at $h = 0$ is an extrapolation, not an additional simulation. Reference lines do not enter the fit.

Three data points determine the three model parameters; a successful fit is therefore not an independent test of the power-law assumption. These are engineering uncertainty estimates, not guaranteed bounds or statistical confidence intervals. The fitted p belongs to this observable and grid range, not to every quantity produced by the solver.

11.2 An independent fourth-grid check at Re = 400

An additional 513^2 run starts from rest and meets the same stopping criterion. Comparing overlapping triples tests the original extrapolation:

N values	Fitted p	Extrapolated $\psi_{\min}$	Fine-grid GCI (%)
65, 129, 257	2.027	-0.1139863	0.327
129, 257, 513	2.021	-0.1139880	0.081

The original three-grid model predicts $\psi_{\min} = -0.1139135$ on 513^2; the independent integration gives -0.1139144, a difference of 9.51×10^{-7}. The two extrapolated limits differ by 1.67×10^{-6}. This is an independent check of the primary-streamfunction model at Re = 400 only. It does not establish the same reliability for other Reynolds numbers or observables.

11.3 Other observables and complete centerline differences

The same procedure is applied to the vortex coordinates, kinetic energy, center vorticity, and bottom-right secondary-vortex strength.

Re	Observable	Fitted p	GCI (%)	Interpretation
400	x_c	2.102	0.052	conditional
400	y_c	2.071	0.056	conditional
400	E_k	1.991	0.512	conditional
400	$\omega(1/2, 1/2)$	2.059	0.102	conditional
400	ψ_{BR}	N/A	N/A	no admissible fit
1000	x_c	1.946	0.073	conditional
1000	y_c	2.176	0.067	conditional
1000	E_k	1.856	1.807	conditional
1000	$\omega(1/2, 1/2)$	1.983	0.770	conditional
1000	ψ_{BR}	2.248	0.534	conditional
3200	x_c	1.721	0.142	conditional
3200	y_c	2.340	0.070	conditional
3200	E_k	1.785	6.695	conditional
3200	$\omega(1/2, 1/2)$	1.921	3.056	conditional
3200	ψ_{BR}	2.260	1.771	conditional

Profile differences use 65 common, uniformly spaced grid nodes, including the endpoints; no interpolation is needed. Their L^2 norm uses composite trapezoidal integration on $[0, 1]$, distinct from the RMS over Ghia's 17 irregularly spaced stations.

Re	Grid pair	$\|\delta u\|_2$	$\|\delta v\|_2$
400	65^2 to 129^2	1.027e-02	1.075e-02
400	129^2 to 257^2	2.520e-03	2.633e-03
1000	65^2 to 129^2	2.528e-02	2.607e-02
1000	129^2 to 257^2	6.480e-03	6.626e-03
3200	129^2 to 193^2	1.528e-02	1.558e-02
3200	193^2 to 257^2	5.535e-03	5.619e-03

At Re = 400, the bottom-right eddy strength is non-monotone across the three grids, so no GCI is assigned. At Re = 3200, the kinetic-energy GCI is 6.695%, versus 3.321% for the primary streamfunction. Position estimates also depend on the local quadratic fit used to locate the extremum.

Interpretation. The main vortex's uncertainty estimate cannot be transferred to secondary eddies or to the whole velocity field.

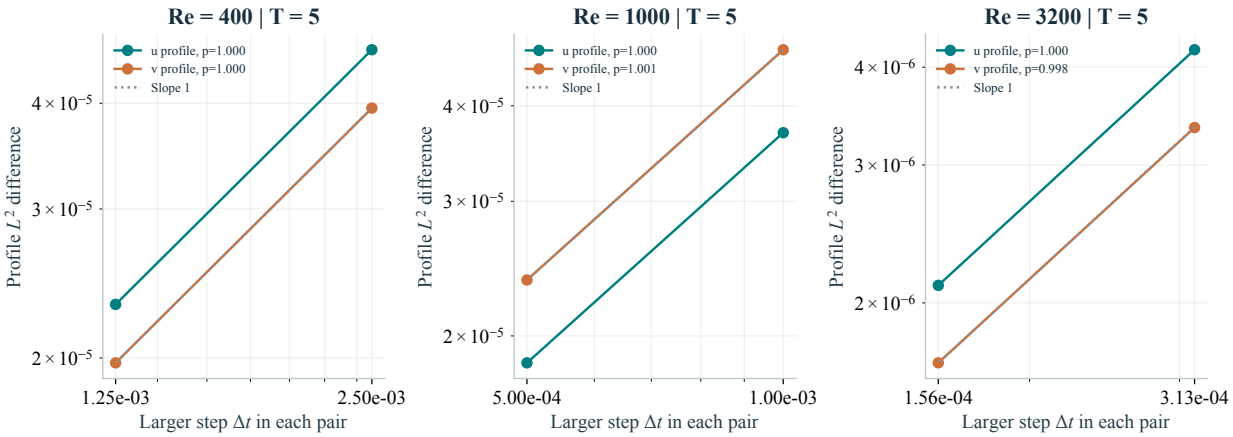
11.4 Time refinement at a common transient time

For each Reynolds number, integrate afresh from rest on the same 129^2 grid to $T = 5$ with Δt, $\Delta t/2$, and $\Delta t/4$. The target is an integer multiple of every step. Define

$$D_u(\Delta t) = \|u_{\Delta t}(1/2, \cdot, T) - u_{\Delta t/2}(1/2, \cdot, T)\|_2, \quad p_{t,u} = \log_2 \frac{D_u(\Delta t)}{D_u(\Delta t/2)}, \tag{33}$$

and similarly for v. Using a common grid isolates temporal self-convergence from spatial refinement. It does not measure the continuum error at that grid.

Re	Base Δt	$D_u(\Delta t)$	$D_u(\Delta t/2)$	$p_{t,u}$	$p_{t,v}$
400	2.500e-03	4.628e-05	2.314e-05	1.000	1.000
1000	1.000e-03	3.688e-05	1.844e-05	1.000	1.001
3200	3.125e-04	4.211e-06	2.106e-06	1.000	0.998



Successive profile differences from actual cavity transients. All runs use the impulsively started, discontinuous unit lid.

11.5 Why a steady solution has a different time-step question

At an exactly stationary discrete state, $\omega^{n+1} = \omega^n = \omega$ and the lagged boundary is also stationary. Subtracting ω from (14) and dividing by Δt gives

$$C_h(\psi, \omega) - \mathrm{Re}^{-1}\Delta_h\omega = 0. \tag{34}$$

Thus the time step cancels out of the fixed-point equations. There is no separate backward-Euler truncation bias in their exactly converged solution. The step still affects transient accuracy, stability, and how closely the iteration approaches that fixed point. Different stable steps can also have different attraction behavior if multiple discrete solutions exist.

This is why the time-order experiment is performed during the transient. For the final states, halving the step is instead used as a continuation sensitivity check, together with an unfiltered steady-equation residual.

11.6 Measure settling instead of relying only on a threshold

Refresh the wall vorticity from the current ψ and evaluate

$$F_h = C_h(\psi, \omega) - \mathrm{Re}^{-1}\Delta_h\omega, \qquad R_{\mathrm{PDE}} = \|F_h\|_\infty. \tag{35}$$

The monitored update rate is related to the old-state defect by $(I + \Delta t \mathrm{Re}^{-1}L)(\omega^{n+1} - \omega^n)/\Delta t = -F_h$. It is therefore a filtered residual, not numerically identical to R_{PDE}.

Starting independently from each saved final state, integrate ten further time units with the original step and with half that step. The following table reports the initial steady defect, the largest monitored update rate across both continuations, and the larger observable drift of the two trials. Halving the step reduces the implicit filtering, so its initial update rate can marginally exceed the original stopping threshold even though the initial fields are identical. These small excursions are retained; the original-step stopping criterion is not retroactively changed.

Re	Grid	R_{PDE}	$R_{\max}$	$\lvert\delta\psi_{\min}\rvert$	$\|\delta u\|_2$	$\|\delta v\|_2$
400	65^2	1.01e-06	9.639e-07	3.00e-08	1.33e-07	1.01e-07
400	129^2	1.08e-06	9.520e-07	1.20e-08	9.09e-08	8.66e-08
400	257^2	1.32e-06	9.756e-07	2.08e-08	9.27e-08	9.12e-08
1000	65^2	1.00e-06	9.972e-07	1.18e-07	2.67e-07	2.42e-07
1000	129^2	1.02e-06	9.990e-07	1.01e-07	2.30e-07	2.10e-07
1000	257^2	1.07e-06	9.951e-07	8.98e-08	2.04e-07	1.87e-07
3200	129^2	1.00e-06	1.001e-06	1.15e-07	2.62e-07	2.49e-07
3200	193^2	1.00e-06	1.002e-06	1.01e-07	2.30e-07	2.19e-07
3200	257^2	1.01e-06	1.002e-06	9.31e-08	2.13e-07	2.03e-07

This measures sensitivity over a specified interval. It is not a proof that all later changes are smaller, and the residual is not divided by an unjustified decay constant to manufacture an error bound.

The fresh fine-grid integrations also save intermediate states. Comparing the former observation times to the converged states quantifies the effect that was previously left unresolved:

Re	Earlier t	$\psi_{\min}$ change (%)	$\|\delta u\|_2$	$\|\delta v\|_2$
1000	150	0.0001	2.257e-07	2.068e-07
3200	300	0.0478	1.296e-04	1.237e-04

Re	Grid GCI (%)	Settling drift (%)	Step-halving $\lvert\delta\psi_{\min}\rvert$
400	0.327	1.830e-05	5.49e-13
1000	0.941	7.603e-05	5.48e-13
3200	3.321	7.843e-05	5.01e-13

The three columns are distinct sensitivity measures on the 257^2 grids, not independent random errors to combine in quadrature. The first assumes a power-law spatial error; the second measures ten-time-unit drift; the last compares the endpoints of the two continuations. The measured settling effects can thus be assessed against spatial sensitivity without treating the residual threshold as an uncertainty estimate. Across these three fine-grid cases, the conditional grid estimate is at least 1.24×10^4 times the measured primary-vortex settling drift. Within these experiments, spatial resolution is the dominant measured limitation for primary-vortex strength. This conclusion does not transfer automatically to every flow quantity.

Interpretation. The completed experiments distinguish measured settling, transient time accuracy, conditional grid uncertainty, and disagreement with a numerical reference. Each answers a different question about the computed flow.

12 Software design and reproduction

Module	Responsibility
grid.py	Coordinates, array layout, spacing and validated dimensions
operators.py	Stencils, sparse matrices, DST/LU/CG, step-size guidance
boundary.py	Four Thom wall formulas and corner storage convention
problems.py	Cavity data, symbolic manufactured forcing, restart signature
solver.py	One time step, monitoring, snapshots and checked restart
postproc.py	Velocity, extrema, quadrature and algebraic diagnostics
benchmark.py	Published data, explicit flags, coordinate transformation
verification.py	Spatial, temporal and joint refinement experiments
accuracy.py	Observable differences, steady defect and conditional grid extrapolation

Checkpoints include the mesh lengths, time step, boundary configuration, and built-in problem parameters. Strict loading rejects incompatible states before changing the solver. An explicit non-strict import is available for continuing an old state under a deliberately changed configuration. Custom problems must include all physical parameters in their checkpoint signature. Restart with DST and the same configuration is tested bit for bit; changing backends only promises agreement within solver tolerance.

12.1 Reproduce the project from its root directory

```
python -m pip install -e ".[dev,notebook]"
python -m pytest -q
python scripts/run_demo.py
python scripts/run_production.py --force
python scripts/check_fourth_grid.py
python scripts/run_accuracy.py
python scripts/build_results.py
python scripts/build_notebook.py --execute
python scripts/build_report.py
```

The demonstration uses a small 65^2 grid. The production suite is much more expensive, especially $\mathrm{Re} = 3200$. Field caches are deliberately excluded from Git; committed figures, tables, measured JSON evidence and the executed notebook remain readable without them. The report build requires Tectonic or a TeX Live installation. See docs/reproducibility.md for partial builds and the precise provenance of saved results.

12.2 Next numerical experiments

The next useful accuracy experiment extends the fourth-grid check from $\mathrm{Re} = 400$ to the higher Reynolds numbers and to wall-sensitive observables. An Arakawa advection option and higher-order time integration with a compatible wall update are separate method extensions. Grid stretching could target boundary layers but would require replacing the present sine-transform diagonalisation. A smoothed lid is useful for a separate regularity study and should be labelled as a different problem.

The current result is a compact, inspectable demonstration of solving, verifying and interpreting the two-dimensional Navier–Stokes equations. Its strongest claims are the ones connected directly to reproducible tests and published comparison data.

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[8] I. B. Celik, U. Ghia, P. J. Roache, C. J. Freitas, H. Coleman and P. E. Raad (2008). Procedure for estimation and reporting of uncertainty due to discretization in CFD applications. *Journal of Fluids Engineering*, 130(7), 078001. doi:10.1115/1.2960953.

Data provenance. The comparison arrays were checked against the original tables. The paper scans are not redistributed in this repository. Source links, flags, and the Botella–Peyret reflection are documented in docs/references.md. The simulation and verification measurements are exported as machine-readable files under data/.

Reading the figures. All flow figures are generated from numerical fields. Streamlines show the instantaneous velocity direction. Vorticity colour limits and local eddy contour normalisations are explicitly stated. No generated illustration is used as evidence of a computed solution.